DataNova: Automated EDA Report

Format: Technical | Domain Detected: Healthcare

Executive / Dataset Overview

The dataset under review captures a snapshot of 309 patient records drawn from a healthcare context focused on lung-cancer risk assessment. It comprises 16 variables, of which one (GENDER) and the target (LUNG_CANCER) are string-type categorical fields, while the remaining 14 are numeric integer indicators of lifestyle, clinical symptoms, or comorbidities. The numeric fields are binary (0/1) except for AGE, which records the patient's age in years. The target variable LUNG_CANCER denotes whether a patient has been diagnosed with lung cancer (presumably "YES"/"NO"). No rows or columns are missing, and the schema is internally consistent with the declared data types.

Data Quality & Profiling Insights

Missing Values - The missing-value audit returned an empty dictionary, confirming that every cell in the 309 × 16 matrix is populated. Consequently, imputation is not required at this stage.

Data-type Consistency - All columns conform to their declared dtypes: GENDER and LUNG_CANCER are object/string, AGE is int64, and the remaining 13 symptom/behavior columns are int64. No dtype mismatches were detected.

Outlier Detection (IQR Method) - AGE - Using the inter-quartile range (IQR) approach, the lower and upper bounds for a non-outlier age were calculated as 39.0 and 87.0, respectively. - Two observations fall outside these bounds: • Row index 22 with AGE = 21 (below lower bound) • Row index 238 with AGE = 38 (below lower bound) - No ages exceed the upper bound of 87.0. All other numeric variables are binary (0/1) and therefore do not generate IQR-based outliers.

Potential Data-Quality Nuances - Column names contain trailing spaces in "FATIGUE ", "ALLERGY ", and "CHRONIC DISEASE". These invisible characters can cause downstream processing errors (e.g., when selecting columns programmatically). - The target variable LUNG_CANCER is stored as a string; its distinct values should be enumerated (e.g., "YES", "NO") to verify consistency and to decide on label encoding.

Strategic / Operational Takeaways

1. Data-Cleaning Actions - Strip leading/trailing whitespace from column names to standardize the schema (e.g., rename "FATIGUE " → "FATIGUE"). - Verify the categorical levels of GENDER and LUNG_CANCER for spelling variations or case inconsistencies; unify them (e.g., "Male", "male", "M" → "Male"). - Address the two age outliers. Options include: a) Capping (winsorizing) ages at the nearest non-outlier bound (39 for values below 39). b) Removing the two rows if they are deemed data entry errors and their exclusion does not jeopardize class balance. c) Retaining them but flagging as "age_outlier" for model robustness checks.

2. Feature Engineering Recommendations - Encode GENDER using binary (0/1) or one-hot encoding, depending on the modeling algorithm. - Convert the target LUNG_CANCER to a binary numeric label (e.g., 1 = YES, 0 = NO) for supervised learning. - Consider aggregating symptom variables into composite scores (e.g., total symptom count) to capture overall clinical burden. - Explore interaction terms between smoking status and age, as age-smoking synergy is clinically relevant for lung-cancer risk.

3. Modeling Guidance - The dataset is modest in size (n = 309) with a relatively high dimensionality (15 predictors). Regularized logistic regression (L1/L2) or tree-based ensembles (Random Forest, Gradient Boosting) are appropriate initial models. - Prior to model training, assess class distribution of LUNG_CANCER. If the positive class is under-represented, apply resampling techniques (SMOTE, stratified undersampling) or adjust class weights. - Perform stratified k-fold cross-validation (k = 5 or 10) to obtain reliable performance estimates given the limited sample size.

4. Validation & Monitoring - Because age outliers exist, include age-related diagnostics (e.g., partial dependence plots) to ensure the model's predictions are not driven by extreme values. - Track model calibration (e.g., reliability diagrams) since clinical decision support demands well-calibrated probability outputs.

5. Operational Deployment Considerations - Implement a preprocessing pipeline that enforces column-name sanitization, categorical encoding, and outlier handling consistently across training and inference environments. - Log any rows flagged as age outliers during inference for auditability and potential manual review. - Establish a feedback loop with clinicians to validate model predictions against real-world outcomes, enabling periodic retraining as new patient data accrue.

In summary, the dataset is clean with respect to missingness and type consistency, but it requires minor preprocessing to resolve whitespace in column names, standardize categorical labels, and decide on a strategy for two low-age outliers. Once these steps are completed, the data are well-positioned for binary classification modeling aimed at predicting lung-cancer risk, with clear pathways for feature engineering, model validation, and operationalization.

Visual Analysis